IoT-Based Smart Toll Collection System using ESP32

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***Abstract*—The increasing volume of vehicular traffic on highways has made traditional toll collection systems inefficient, leading to long queues, fuel wastage, and increased operational delays. To address these challenges, this study proposes an IoT-Based Smart Toll Collection System using ESP32 to enable automated, contactless, and efficient toll processing. The system integrates RFID-based vehicle identification, IR sensor for vehicle detection, and cloud connectivity through Wi-Fi-enabled ESP32 modules to ensure real-time toll management. Each vehicle is assigned a unique RFID tag, which is detected as it approaches the toll plaza. Upon successful authentication, the system automatically deducts the toll fee from a pre-registered account and updates the transaction data to a cloud database. IR sensors are used to verify vehicle presence, ensuring accurate detection. The ESP32 microcontroller acts as the central processing unit, coordinating sensor input, network communication, and gate control operations. The system also includes a barrier control mechanism that opens only after successful payment validation, ensuring security and preventing unauthorized passage. Real-time monitoring is enabled through a web dashboard, allowing authorities to track transactions, vehicle logs, and system status efficiently. In addition, failure cases such as insufficient balance or invalid RFID trigger alerts and deny gate access. In general, the proposed smart toll collection system improves efficiency, reduces congestion, minimizes human intervention, and provides a scalable and cost-effective solution for modern transportation infrastructure.**

***Index Terms*—IoT, Smart Toll System, ESP32, RFID, Automation, Real-time Monitoring, Microcontroller, Payment Automation, Highway Management, IoT, Smart Toll System, ESP32, RFID, Automation, Cloud Computing, Traffic Management.**

Identify applicable funding agency here. If none, delete this.

I. Introduction

The rapid increase in vehicle population and highway usage has made efficient toll collection a major concern in modern transportation systems. Traditional manual toll booths often lead to long queues, delays, fuel wastage, and operational inefficiency. As traffic density continues to grow, there is a strong need for automated and intelligent tolling solutions that can ensure faster processing and improved accuracy.

The motivation of this study is to address the limitations of existing toll collection methods and to develop a low-cost, automated system that reduces human intervention. In many regions, including developing countries, toll plazas still rely on semi-manual operations, which create congestion and reduce overall system efficiency.

The Internet of Things (IoT) plays a key role in modern transportation systems by enabling real-time communication between devices, sensors, and cloud platforms. In toll collection systems, IoT technologies such as RFID, ESP32 microcontrollers, and wireless connectivity allow automatic vehicle identification, real-time processing, and remote monitoring, thereby improving efficiency and transparency.

However, existing systems still face several challenges, including limited multi-sensor integration, lack of real-time cloud monitoring, scalability issues, and occasional false detection in vehicle identification. These gaps highlight the need for a more reliable and integrated smart tolling framework.

To overcome these issues, this paper proposes an IoT-Based Smart Toll Collection System using ESP32. The system integrates RFID-based vehicle identification, load sensor-based

detection, automated gate control, and cloud connectivity for real-time monitoring and payment processing. The main contribution of this work is a unified and scalable architecture that improves toll efficiency, reduces congestion, and minimizes manual dependency.

The remainder of this paper is organized as follows: Section II presents the literature review, Section III describes the system design and methodology, Section IV discusses the implementation and results, and Section V concludes the paper with future scope.

II. LITERATURE REVIEW

The development of automated toll collection systems has been widely explored in the field of Intelligent Transportation Systems (ITS) to reduce congestion, improve efficiency, and minimize human intervention. This section reviews key contributions from previous research relevant to IoT-based smart toll systems.

RFID-based toll collection systems are among the earliest and most widely studied approaches for automation. Shridevi Soma et al. [1] proposed an IoT-based toll collection system using RFID technology, where vehicles are identified through RFID tags and toll charges are automatically deducted. Their study demonstrated significant reduction in manual processing time and improved throughput at toll booths. However, the system lacked advanced real-time monitoring and cloud-based analytics.

Similarly, Kavyashree M et al. [2] developed a smart toll collection system using RFID, focusing on improving vehicle identification accuracy and reducing congestion. Their research highlighted that RFID-based automation can significantly reduce waiting time compared to manual toll collection. However, challenges such as tag misreading and limited scalability in high-traffic environments were identified.

Yashodhre K et al. [3] extended the concept by integrating IoT technologies into toll booth management systems. Their work introduced remote monitoring capabilities and improved data handling using network connectivity. The study emphasized that IoT integration enhances system responsiveness, but issues related to real-time synchronization and network dependency remained limitations.

In another approach, Gupta et al. [5] proposed a smart toll system using ZigBee and RFID technologies. Their system enabled wireless communication between toll components, improving operational flexibility. The study concluded that wireless sensor networks can enhance automation but require careful optimization for reliability and latency.

Brindha et al. [6] developed an IoT-based electronic toll collection system combining RFID and cloud connectivity. Their work showed that integrating IoT platforms enables better scalability and centralized monitoring of toll transactions. However, they noted that security and authentication mechanisms need further improvement to prevent data manipulation.

Recent works such as Tiwari et al. [4] introduced ESP32-based IoT toll systems that leverage Wi-Fi connectivity for real-time data transmission and automated control. Their findings indicate that ESP32 microcontrollers provide a cost-effective and efficient solution for smart toll infrastructure due to their built-in wireless capabilities and low power consumption.

Previous toll systems mainly used manual collection and later RFID-based automation [1], [2], which improved speed but still lacked real-time monitoring, scalability, and strong security. IoT-based and wireless systems [5], [7] added remote access, but often suffered from network dependency and incomplete integration. Some ESP32-based solutions [4] improved connectivity, but still did not combine all functions into a unified system.

The proposed system improves these approaches by combining RFID and IR sensors for accurate vehicle detection, using ESP32 for real-time cloud connectivity, and enabling automated gate control based on multi-level verification, resulting in a more reliable, efficient, and scalable toll collection system.

III. PROPOSED SYSTEM/ARCHITECTURE

A. System Overview

The proposed IoT-Based Smart Toll Collection System using ESP32 is designed to automate toll processing through real-time vehicle identification, verification, and gate control. The ESP32 microcontroller serves as the central unit that integrates sensor data, controls the barrier mechanism, and manages wireless communication with a cloud platform for real-time data logging and monitoring.

The system uses an RFID module for vehicle identification, a load sensor (or IR sensor) for detecting vehicle presence, and a servo motor or barrier gate for access control. When a vehicle arrives at the toll plaza, its RFID tag is scanned and verified against a registered database. If the vehicle is authorized and sufficient balance is available, the toll fee is automatically deducted and the gate is opened. Otherwise, access is denied and the barrier remains closed. All transactions are updated in real time to a cloud server for monitoring and record keeping.

TABLE I
FEATURE COMPARISON TABLE

Feature	**Traditional Toll System**	**Proposed IoT-Based System (ESP32)**
Vehicle Identification	Manual ticket/check	RFID-based automatic identification
Payment Method	Cash-based	Automated digital deduction
Processing Time	High (delays/queues)	Low (real-time processing)
Human Intervention	High	Minimal
Error Rate	Moderate to high	Low due to sensor + RFID validation
Data Recording	Manual logs	Cloud-based real-time logging
Monitoring	Local only	Remote web dashboard
Scalability	Limited	High and expandable
Traffic Flow	Often congested	Smooth and continuous flow

Fig. 1. System workflow.

B. Hardware Components

The proposed IoT-Based Smart Toll Collection System using ESP32 is implemented using a set of embedded hardware modules that work together to enable automated toll processing, vehicle detection, and real-time communication. The key hardware components of the system are described below:

- **ESP32 Microcontroller:** The ESP32 serves as the core processing unit of the system. It handles RFID data processing, sensor integration, Wi-Fi communication, and control of the gate mechanism. Its built-in Wi-Fi capability enables seamless connection to cloud platforms for real-time monitoring and data logging.
- **RFID Module (RC522):** The RFID reader is used to identify vehicles equipped with RFID tags. Each tag contains a unique ID linked to a user account. When a vehicle approaches the toll booth, the RFID module scans the tag and sends the data to the ESP32 for authentication.
- **RFID Tags/Cards:** These are assigned to individual vehicles. Each tag is pre-registered in the system database with user details and account balance, enabling automatic toll deduction upon successful verification.
- **IR Sensors:** These sensors detect the presence of a vehicle at the toll point. It ensures that the system only processes transactions when a vehicle is physically present, thereby reducing false readings and improving system reliability.
- **Servo Motor / Barrier Gate:** The servo motor controls the opening and closing of the toll gate. Upon successful authentication and payment confirmation, the ESP32 sends a signal to the servo motor to lift the barrier, allowing the vehicle to pass.
- **OLED Display:** A display unit can be used to show transaction details such as vehicle ID, balance, and toll status (e.g., “Access Granted” or “Insufficient Balance”).
- **Buzzer:** These components provide user feedback. Buzzer indicates successful payment, errors, or unauthorized access attempts.

C. Software and Communication Framework

The proposed system utilizes a lightweight and efficient software architecture combining embedded programming, cloud scripting, and web technologies. The ESP32 is programmed using Arduino IDE, where the core logic for RFID authentication, sensor handling, and gate control is implemented. For backend processing and data management, Google Apps Script is used to handle HTTP requests, process transaction data, and store records in Google Sheets, providing a simple and cost-effective cloud solution.

The application layer is developed using HTML and CSS, creating a web-based interface for monitoring toll transactions, vehicle data, and system status. This dashboard allows administrators to view real-time updates and manage records remotely.

In terms of communication, the ESP32 connects to the internet via Wi-Fi and sends data to the Google Apps Script web app using HTTP/HTTPS requests. The script processes the incoming data and updates the database accordingly. The RFID module communicates with the ESP32 using SPI protocol, while sensors and actuators interact through GPIO pins.

D. Data Acquisition and Processing

The proposed IoT-Based Smart Toll Collection System follows a structured approach for data collection, transmission, storage, and processing to ensure accurate and real-time toll operations.

Data Acquisition: Data is primarily collected through the RFID module and vehicle detection sensors. When a vehicle approaches the toll point, the RFID reader scans the tag and retrieves a unique identification number. Simultaneously, the IR sensors detect the presence of the vehicle, ensuring that only valid entries are processed. This dual-input mechanism improves system accuracy and reduces false detections.

Data Transmission: The collected data, including RFID ID and sensor status, is transmitted by the ESP32 to a cloud-based backend using Wi-Fi. HTTP/HTTPS requests are sent to a Google Apps Script web service, which acts as an intermediary for processing and storing the data.

Data Storage: The Google Apps Script stores incoming data in Google Sheets, which serves as a lightweight cloud database. Each transaction, including vehicle ID, timestamp, and toll status, is recorded for future reference and monitoring.

Data Processing: Upon receiving the data, the Apps Script validates the RFID ID, checks the associated account balance, and determines whether the transaction is successful. Based on this decision, a response is sent back to the ESP32 to either open the gate via servo motor. The system also updates the balance and logs the transaction in real time.

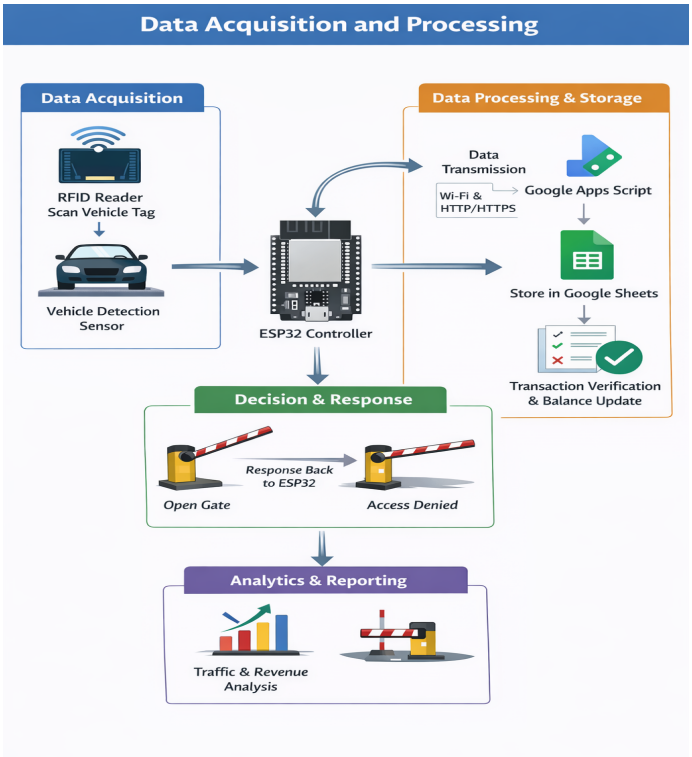

Fig. 2. Data acquisition and processing workflow.

E. Automation and Actuation

The IoT-based smart toll system uses sensors, smart logic, and automatic controls for efficient, contactless toll collection.

Automation Mechanism: The automation process begins when a vehicle approaches the toll point and is detected by the IR sensors. This triggers the ESP32 to activate the RFID reader for scanning the vehicle's RFID tag. Once the tag is read, the ESP32 processes the data and sends it to the backend (Google Apps Script) for verification, including user authentication and balance checking.

Actuation Process: The actuation stage involves controlling physical devices based on the processed data. If the transaction is successful, the ESP32 sends a signal to the servo motor, which lifts the barrier gate to allow the vehicle to pass. After a short delay, once the vehicle leaves the detection zone, the gate automatically closes. In case of authentication failure or insufficient balance, the system denies access by keeping the gate closed. Additionally, OLED display and buzzer trigger to notify the user of the transaction status (e.g., success or error).

F. Security and Privacy Considerations

Data exchanged between the ESP32 and the backend (Google Apps Script) is transmitted using HTTPS, which provides SSL/TLS encryption to protect sensitive information such as RFID IDs and transaction details from interception. Authentication is implemented by validating RFID tags against a registered database, ensuring that only authorized vehicles can access the system. Data stored in Google Sheets is protected access control permissions, allowing only admins to view or modify records.

IV. Prototype Implementation and Experimental Setup

A working prototype of the IoT-Based Smart Toll Collection System was developed using ESP32, RFID module (RC522), IR sensors, a buzzer and a servo motor for gate control. The ESP32 was programmed to handle RFID authentication, sensor input, and communication with a cloud backend implemented using Google Apps Script and Google Sheets. A simple web interface built with HTML and CSS was used for real-time monitoring of transactions and system status. In operation, the sensor detects a vehicle, the RFID tag is scanned, and the ESP32 verifies the data through the cloud. If valid, the toll is deducted and the gate opens; otherwise, access is denied, and the event is logged.

The system was tested by using multiple RFID tags to represent different vehicles. A stable Wi-Fi connection ensured continuous communication with the cloud. Since no machine learning was used, the evaluation relied on real-time data generated during testing.

The system achieved high accuracy in identifying valid tags, low latency in gate operation, and consistent performance under repeated use. These metrics were chosen to reflect real-world requirements such as fast processing, dependable operation, and low power consumption, demonstrating the practicality of the proposed system.

Fig. 3. System workflow.

V. Results and Discussion

A. Experimental Results

The system was tested under multiple vehicle scenarios to evaluate accuracy, response time, and reliability. The system successfully identified RFID tags, verified account balance, and controlled the gate operation in real time. The average processing time per vehicle was observed to be significantly low (within a few seconds), ensuring smooth traffic flow. The integration of sensor-based detection minimized false triggering, and all transactions were accurately recorded in Google Sheets via Google Apps Script. The system demonstrated high reliability in handling consecutive vehicle entries, with consistent communication between ESP32 and the backend. Cases of invalid RFID or insufficient balance were correctly handled by denying access, confirming the effectiveness of the decision-making logic.

B. Comparative Analysis

The system shows significant improvements in processing speed, automation, and real-time monitoring compared to traditional systems. It reduces congestion, eliminates manual errors, and provides transparent digital records. However, some limitations exist, such as dependency on stable internet connectivity and potential delays in cloud response under poor network conditions. Additionally, security and scalability can be further enhanced with advanced encryption and database systems in future implementations. Overall, the system demonstrates efficient performance and provides a practical, low-cost solution for smart toll management.

TABLE II
Comparison Table of Traditional System and Proposed System

Parameter	Traditional System	Proposed System
Processing Time	High	Low
Human Intervention	High	Minimal
Accuracy	Moderate	Very High (sensor + RFID)
Real-time Monitoring	No	Yes (cloud-based)
Cost	Moderate	Low-cost (ESP32-based)

VI. Conclusion and Future Work

This paper presented an IoT-Based Smart Toll Collection System using ESP32 that automates vehicle identification, toll deduction, and gate control. By integrating RFID technology, sensor-based detection, and cloud connectivity through Google Apps Script, the system ensures fast, accurate, and contactless toll processing. The proposed solution reduces traffic congestion, minimizes human intervention, and provides real-time monitoring, making it a cost-effective and scalable approach for modern toll management.

Future improvements can focus on enhancing system security through advanced encryption and authentication methods. Integration of mobile payment systems and digital wallets can further improve user convenience. Additionally, implementing machine learning techniques for traffic prediction and congestion analysis could optimize toll operations. Expanding the system with multi-lane support, offline data handling, and a more robust cloud database can further increase reliability and scalability.

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